Nanotechnology

A **nanoparticle** is tiny. It is a **molecule** or part molecule and is sometimes called a nanomolecule. They can travel easily through blood vessels and enter some cells. Scientists are trying to find ways to use these tiny chemical particles in medicine. Possible uses could be:

- coating a medicine with nanoparticles so it goes only to the organ where it is needed
- attaching nanoparticles to a medicine so it can cross through a body barrier, such as from blood into the brain
- to give small doses of medicine that isn't wasted in other body parts because it only goes where it is needed
- protecting a vaccine so it can get into the cells where it can work
- delivering dangerous chemotherapy drugs directly to the cancer cells only
- finding damaged cells and repairing them
- attaching nanoparticles to a tumour to make it visible to imaging machines
- making purer medicines to reduce side effects caused by impurities.

Nanotechnology requires many years more research. Because the nanoparticles are so small and not enough is known about them yet, there are possible dangers. Will they stay in the body long after their use is over? Will they stay unchanged in the body or could they change and cause problems? Will they get into the environment and cause pollution problems? These possible dangers are studied in **nanotoxicology**.

☞ **1** Replace the missing vowels, then join the sentence halves to make complete sentences.

Nan__p__rt__cl__s may find	against potential d__ng__rs of nanoparticles.
Nanoparticles may __tt__ch to	to make s__r__ they go to their t__rg__t.
N__n__m__l__c__l__s may coat dr__gs	t__m__ __rs and help with early d__ __gn__s__s.
Nanoparticles may __ll__w us to use	d__m__g__d cells and f__x them.
N__n__t__x__c__l__gy is trying to g__ __rd	much smaller am__ __nts of m__d__c__n__s.

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